

Shinji Watanabe

Associate Professor, Language Technologies Institute, Carnegie Mellon University

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Google Scholar · GitHub · WAVLab · CMU LTI

EDUCATION

1. Ph.D. from Waseda University, School of Science and Engineering, February 2006, Supervisors: Profs. T. Kobayashi, Y. Matsuyama, Y. Sagisaka, and T. Matsushima
2. Master of Science from Waseda University, March 2001, Supervisor: Prof. Ichiro Ohba
3. Bachelor of Science from Waseda University, March 1999, Supervisor: Prof. Ichiro Ohba

APPOINTMENTS

1. January 2021 - present: "Associate Professor," Language Technologies Institute at Carnegie Mellon University, Pittsburgh PA, USA
2. April 2025 - present: "Visiting Professor," Center for Language AI Research, Tohoku University, Japan
3. April 2025 - present: "Visiting Researcher," NHK Science & Technology Research Laboratories, Japan
4. September 2022 - present: "Courtesy Faculty," Department of Electrical and Computer Engineering at Carnegie Mellon University, Pittsburgh PA, USA
5. January 2021 - 2025: "Adjunct Associate Professor," Center for Language and Speech Processing at John Hopkins University, Baltimore MD, USA
6. July 2017 - December 2020: "Associate Research Professor," Department of Electrical and Computer Engineering at John Hopkins University, Baltimore MD, USA
7. October 2014 - June 2017: "Senior Principal Research Scientist" (formerly, Senior Principal Member Research Staff) at Mitsubishi Electric Research Laboratories (MERL), Cambridge MA, USA (Group manager: Dr. Anthony Vetro. Team leader: Dr. John R. Hershey)
8. April 2013 - September 2014: "Principal Member Research Staff" at Mitsubishi Electric Research Laboratories (MERL), Cambridge MA, USA (Group manager: Dr. Anthony Vetro. Team leader: Dr. John R. Hershey)
9. January 2012 - March 2013: "Visiting Member Research Staff" at Mitsubishi Electric Research Laboratories (MERL), Cambridge MA, USA (Group manager: Dr. Anthony Vetro. Team leader: Dr. John R. Hershey)
10. July 2011 - November 2011: "Research Scientist" at Nippon Telegraph and Telephone (NTT) Communication Science Laboratories, Signal Processing Research Group, Kyoto, Japan (Group leader: Dr. Atsushi Nakamura)
11. January 2009 - March 2009: "Visiting Scholar" at Georgia Institute of Technology, Atlanta, USA. (Supervisor: Professor Biing-Hwang (Fred) Juang)
12. April 2001 - June 2011: "Researcher" at Nippon Telegraph and Telephone (NTT) Communication Science Laboratories, Signal Processing Research Group, Kyoto, Japan (Group leader: Dr. Shoji Makino → Dr. Masato Miyoshi → Dr. Atsushi Nakamura)

AWARDS AND HONOURS

1. Co-authored paper achieved the IEEE ICASSP Best Student Paper Award in 2026
2. Co-authored paper achieved the IEEE SPS Young Author Best Paper Award in 2025
3. Co-authored paper achieved the ISCA Interspeech Best Student Paper Award in 2025
4. Best Paper Award at IEEE SLT in 2024
5. EMNLP Best Paper Award in 2024
6. Judges' Award at the DCASE 2024 Challenge Task 6
7. ISCA Interspeech Best Paper Award in 2024
8. Ranked 1st place at the DCASE 2024 Challenge Task 6

9. Outstanding Reviewer Recognition at IEEE ICASSP in 2023
10. Co-authored paper achieved the IEEE Ganesh N. Ramaswamy Memorial Student Grant (2023)
11. Ranked 1st place at the DCASE 2023 Challenge Task 6A
12. 6 papers are recognized as one of the top 3% of all papers accepted at IEEE ICASSP in 2023
13. Ranked 1st place at the STOP challenge tasks 1 and 3 (2023)
14. Best student paper at IEEE SLT in 2022
15. Best paper candidates (2 papers) at IEEE SLT in 2022
16. Ranked 1st place at The 2nd Clarity Enhancement Challenge (2022)
17. Computer Speech & Language Best Review Paper Award (2022)
18. Ranked 1st place at the IWSLT 2022 Dialect Speech Translation task (2022)
19. Ranked 1st place at the L3DAS22 challenge task 1 (2022)
20. Computer Speech & Language Best Review Paper Award (2021)
21. Ranked 2nd place at The DIHARD III challenge task 1 & 2 full set in 2021
22. Ranked 1st place at The DCASE 2020 challenge task 4 in 2020
23. Facebook Research Award (Towards On-Device AI) in 2020
24. Tencent AI Lab Rhino-Bird Gift Fund in 2020
25. ASRU best paper award at the IEEE ASRU in 2019
26. ASRU best paper candidate (2 papers) at the IEEE ASRU in 2019
27. Google Faculty Research Award in 2019
28. Tencent AI Lab Rhino-Bird Gift Fund in 2018
29. Facebook Research Award (Speech and Audio Technology for Voice Interaction and Video Understanding) in 2018
30. Co-authored paper achieved the ISCA Interspeech Best Student Paper Award in 2018
31. Ranked 1st place at The First DIHARD Speech Diarization Challenge track1 in 2018
32. Best paper candidates (2 papers) and best student paper candidate at the IEEE ASRU in 2017
33. Ranked 3rd place at 4th CHiME Challenge 6 channel track in 2016
34. Ranked 2nd place at 5th Dialog State Tracking Challenge (DSTC5) in 2016
35. Ranked 2nd place at 3rd CHiME Challenge in 2015
36. Ranked 1st place at REVERB Challenge Official Data Track (2nd overall) in 2014
37. Ranked 1st place at 2nd CHiME Challenge Track 2 in 2013
38. Co-authored paper achieved the IEEE ICASSP 2012 Student Paper Award in 2012
39. Best In-Category Nominee at the International Semantic Web Conference in 2011
40. Ranked 1st place at 1st CHiME Challenge in 2011
41. Co-authored paper achieved the IEEE Signal Processing Society Japan Chapter Student Paper Award in 2010
42. The TELECOM System Technology Award from the Telecommunications Advancement Foundation in 2006
43. The Itakura Award from the Acoustical Society of Japan (ASJ) in 2006
44. The Best Paper Award from the Institute of Electronics, Information, and Communication Engineers of Japan (IEICE) in 2004
45. The Awaya Award from the Acoustical Society of Japan (ASJ) in 2003

TEACHING

1. "11752 Speech Generation," Carnegie Mellon University, USA, Fall 2026 with Carlos Busso
2. "11751/18781 Speech Recognition and Understanding," Carnegie Mellon University, USA, Fall 2022--2025
3. "11492/11692/18495: Speech Technology for Conversational AI" Carnegie Mellon University, USA, Spring 2024
4. "11492/11692/18495: Speech Processing" Carnegie Mellon University, USA, Spring 2023
5. "11737: Multilingual Natural Language Processing," Carnegie Mellon University, USA, Spring 2022 with Graham Neubig and Alan W Black

6. "11751/18781 Speech Recognition and Understanding," Carnegie Mellon University, USA, Fall 2021 with Ian Lane
7. "DSTA: Multilingual Natural Language Processing," Carnegie Mellon University, USA, Spring 2021 with Graham Neubig and Alan W Black
8. "EN. 601.667: Introduction to Human Language Technology", Johns Hopkins University, USA, Fall 2019, 2020 with Philipp Koehn and other CLSP faculties
9. "End-to-end speech recognition" JHU Summer School on Human Language Technology (June 20, 2018) with Takaaki Hori
10. "EN. 520.666: Information Extraction", Johns Hopkins University, USA, Spring 2018--2020
11. "Machine Learning for Speech Processing," Department of Electronic Engineering, Tsinghua University, China, December 2012

SKILLS

PROGRAMMING SKILLS

1. Experienced in Linux programming based on C/C++, Python, Matlab, and Bash/Csh
2. Experienced in speech recognition toolkits (Kaldi, ESPnet, HTK, and NTT SOLON) and deep learning toolkits (Theano, Chainer, PyTorch, and CNTK)

LANGUAGE SKILLS

1. Japanese (native)
2. English (fluent)

INVITED TALKS

KEYNOTE TALK

1. **Shinji Watanabe**, "Scaling Multilingual Speech Recognition: From a Handful to Thousands of Languages," *2nd MBZUAI Speech & NLP Symposium* (2025)
2. **Shinji Watanabe**, "Scaling Multilingual Speech Recognition: From a Handful to Thousands of Languages," *Multilingual Conversational Speech Language Modeling (MLC-SLM)* (2025)
3. **Shinji Watanabe**, "Reproducing Large Speech Foundation Models," *Eighth International Workshop on Symbolic-Neural Learning (SNL)*, Tokyo, Japan (2024)
4. **Shinji Watanabe**, "Unifying Speech Processing Applications with Speech Foundation Models," *IEEE Automatic Speech Recognition and Understanding (ASRU)*, Taipei, Taiwan (2023)
5. **Shinji Watanabe**, "Explainable neural network for spoken language processing," *iSpeech 2023*, Jesraem, Israel (2023)
6. **Shinji Watanabe**, "Simplifying Automatic Speech Recognition with Non-Autoregressive Neural End-to-End Modeling," *2020 Accented English Speech Recognition Challenge Workshop (2020 AESRC Workshop)* (2020)
7. **Shinji Watanabe**, "Tackling Multispeaker Conversation Processing based on Speaker Diarization and Multispeaker Speech Recognition," *The VoxSRC Workshop* (2020)
8. **Shinji Watanabe**, "End-to-End Speech Processing: From Pipeline to Integrated Architecture," *Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC) 2018*, Honolulu, Hawaii, USA (2018)
9. **Shinji Watanabe**, "Neural End-to-End Architectures for Speech Recognition in Adverse Environments," *ITG Conference on Speech Communication*, Oldenburg, Germany (2018)
10. **Shinji Watanabe**, "Neural End-to-End Architectures for Speech Recognition in Adverse Environments," *30th annual Conference on Computational Linguistics and Speech Processing (ROCLING)*, Hsinchu, Taiwan (2018)

TUTORIAL/OVERVIEW/INVITED TALK

1. **Shinji Watanabe**, "Open Whisper-Style Speech Models: Transparency, Scalability, and Advancing Explainability," *International Advanced Summer School on NLP (IASNLP)* (2025)
2. Tae Jin Park, Huck Yang, Kyu J. Han, and **Shinji Watanabe**, "Beyond End-to-End ASR: Integrating Long-Context Acoustic and Linguistic Insights," *Interspeech'25* (2025)
3. **Shinji Watanabe**, "Survey Talk: Toward Speech and Audio Foundation Models," *Interspeech'24* (2024)

4. Yossi Adi, Soumi Maiti, and **Shinji Watanabe**, "Recent Advances in Speech Language Models," *Interspeech'24* (2024)
5. Shi-Xiong Zhang , Yong Xu, **Shinji Watanabe**, and Dong Yu, "Recent Advances in Speech Processing, Multi-talker ASR and Diarization for Cocktail Party Problem," *Interspeech'23* (2023)
6. **Shinji Watanabe**, "Explainable End-to-End Neural Networks for Far-Field Conversation Recognition," *SANE 2022 - Speech and Audio in the Northeast* (2022)
7. Abdelrahman Mohamed, Hung-yi Lee, **Shinji Watanabe**, Tara Sainath, Karen Livescu, Shang-Wen Li, Shu-wen Yang, and Katrin Kirchhoff, "Self-supervised Representation Learning for Speech Processing," *Interspeech'22* (2022)
8. Abdelrahman Mohamed, Hung-yi Lee, **Shinji Watanabe**, Tara Sainath, Karen Livescu, Shang-Wen Li, Shu-wen Yang, and Katrin Kirchhoff, "Self-supervised Representation Learning for Speech Processing," *NAACL'22* (2022)
9. Abdelrahman Mohamed, Hung-yi Lee, **Shinji Watanabe**, Tara Sainath, Karen Livescu, Shang-Wen Li, Shu-wen Yang, and Katrin Kirchhoff, "Self-supervised Representation Learning for Speech Processing," *ICASSP'22* (2022)
10. **Shinji Watanabe**, "Machine listening with Explainable AI," *IV 2021 Workshop on Explainable AI on Autonomous Driving* (2021)
11. **Shinji Watanabe**, "Toward Unification of Various Speech Processing Applications based on End-to-End Neural Networks," *Otogaku Symposium* (2021)
12. **Shinji Watanabe**, Pengcheng Guo, and Sathvik Udupa, "Introduction of ESPnet, End-to-End Speech Processing Toolkit," *MULTilingual and Code-Switching ASR Challenges for Low Resource Indian Languages (MUCS)* (2021)
13. Keisuke Kinoshita, Yusuke Fujita, Naoyuki Kanda, and **Shinji Watanabe**, "T-9 - Distant conversational speech recognition and analysis: Recent advances, and trends towards end-to-end optimization," *ICASSP'21* (2021)
14. Takaaki Hori Tomoki Hayashi, Shigeki Karita, and **Shinji Watanabe**, "Advanced Methods for Neural End-to-End Speech Processing - Unification, Integration, and Implementation," *Interspeech'19* (2019)
15. **Shinji Watanabe**, "Hybrid CTC/Attention Architecture for End-to-End Speech Recognition in Challenging Environments," *INTERNATIONAL WORKSHOP ON SPEECH PROCESSING IN CHALLENGING ENVIRONMENTS (SPICE)* (2018)
16. **Shinji Watanabe**, "Multi-microphone Speech Recognition --Recent Advances in Distant Speech Recognition," *Speech and Language Symposium* (2016)
17. **Shinji Watanabe**, Xiong Xiao, and Marc Delcroix "Multi-microphone Speech Recognition," *APSIPA ASC'16* (2016)
18. **Shinji Watanabe**, "Pushing the envelope at both ends: beamforming acoustic models and joint CTC/Attention schemes for end-to-end ASR," *SANE 2016 - Speech and Audio in the Northeast* (2016)
19. Marc Delcroix and **Shinji Watanabe**, "Recent Advances in Distant Speech Recognition," *Interspeech'16* (2016)
20. **Shinji Watanabe** and Jen-Tzung Chien, " Bayesian Learning for Speech and Language Processing," *ICASSP'12 2012, Tutorial T-10*, (2012)
21. **Shinji Watanabe**, "Bayesian approaches in speech recognition," *APSIPA ASC 2011, Plenary Overview Sessions*, (2011)
22. **Shinji Watanabe**, "Recent Topics in Acoustic Modeling for Speech recognition: A Machine Learning Perspective based on Generative and Discriminative Approaches," *IEICE Technical Report, SP2011-31*, pp. 7–10 (2011.6)
23. **Shinji Watanabe**, "Recent topics in speech research," *Tutorial Session 1-2, the 2nd Young Researchers Forum on ALAGIN Speech Processing Session* (2011.3)
24. **Shinji Watanabe**, "Bayesian training of acoustic models," *IPJS SIG Technical Report, Vol.2009-SLP-77, No. 9*, (2009. 7)
25. **Shinji Watanabe**, "Speech recognition based on a Bayesian approach and its application," *Japan Techno Center, Lecture on "Speech recognition technology –application and future trend"* (2007.2.)
26. **Shinji Watanabe**, Yasuhiro Minami, Atsushi Nakamura, and Naonori Ueda, "Speech recognition using variational Bayes," *2005 Workshop on Information-Based Induction Sciences (IBIS2005)*, pp. 269–274, (2005.11)
27. **Shinji Watanabe**, "Selection of shared-state hidden Markov model structure using Bayesian criterion," *IEICE Technical Report, SP2004-149*, pp. 25–30 (2005)
28. **Shinji Watanabe**, "Speech recognition based on the Bayesian approach", *Tutorial lecture on IEICE Technical Report, SP2004-74*, pp. 13--20, (2004)
29. **Shinji Watanabe**, "VBEC: robust speech recognition based on Bayesian approach," *5th young researcher workshop, ASJ Kansai section, I-2*, (2003)

SEMINAR

1. "Scaling Multilingual Speech Recognition: From a Handful to Thousands of Languages," University of Southern California (host: Prof. Shri Narayanan), October 2025
2. "Case Study: How to Manage the Study of Foundation Models in Academia?" Nuts & Bolts Session at TTIC summer workshop on Foundations of Speech and Audio Foundation Models, September 2025
3. "Scaling Multilingual Speech Recognition: From a Handful to Thousands of Languages," Tohoku University (host: Prof. Jun Suzuki), July 2025
4. "Open Whisper-Style Speech Models: Transparency, Scalability, and Advancing Explainability," NTT Communication Science Laboratories (host: Dr. Marc Delcroix), July 2025
5. "Open Whisper-Style Speech Models: Transparency, Scalability, and Advancing Explainability," Kyoto University (host: Prof. Tatsuya Kawahara), June 2025
6. "Open Whisper-Style Speech Models: Transparency, Scalability, and Advancing Explainability," INESC-ID (host: Prof. Alberto Abad), June 2025
7. "Towards a robustly-evaluated and open ecosystem of audio foundation models," with Chris Donahue, Sony AI (host: Dr. Yuki Mitsufuji), May 2025
8. "Open Whisper-Style Speech Models: Transparency, Scalability, and Advancing Explainability," TTIC guest lecture (host: Prof. Karen Livescu), May 2025
9. "Open Whisper-Style Speech Models: Transparency, Scalability, and Advancing Explainability," Workshop on NL and Interactive Systems at Apple, May 2025
10. "Open Whisper-Style Speech Models: Transparency, Scalability, and Advancing Explainability," UC Berkeley (host: Prof. Gopala K. Anumanchipalli), May 2025
11. "Open Whisper-Style Speech Models: Transparency, Scalability, and Advancing Explainability," Conversational AI Reading Group at Mila, February 2025
12. "Speech Self-Supervised Learning Using One Million Hours of Multilingual Data Across Four Thousand Languages," NLP Colloquium, Japan, January 2025.
13. "Reproducing Large Speech Foundation Models," Ohio State University (host: Prof. Deliang Wang)
14. "Reproducing Large Speech Foundation Models," Chulalongkorn University, August 2024 (host: Prof. Ekapol Chuangsuwanich)
15. "Unifying Speech Processing Applications with Speech Foundation Models," DSTA Faculty Speaker Series, July 2024
16. "Toward Explainable Speech Foundation Models," The University of Rochester ECE Distinguished Lecture Series, April 2024 (host: Prof. Zhiyao Duan)
17. "Toward Explainable Speech Foundation Models," invited speaker at the IEEE ICASSP 2023 workshop Explainable Machine Learning for Speech and Audio, April 2024
18. "Unifying Speech Processing Applications with Speech Foundation Models," Amazon, January 2024 (host: Dr. Jin Liu)
19. "Explainable End-to-End Neural Networks for Far-Field Conversation Recognition," Academia Sinica, Taiwan, December 2023 (host: Prof. Yu Tsao)
20. "Variational Bayesian Learning," invited speaker at the Symposium for Celebrating 40 Years of Bayesian Learning in Speech and Language Processing and Beyond, December 2023
21. "Attempts to reproduce large pre-trained speech models on an academic computing scale," Apple, September 2023 (host: Dr. Ahmed Abdelaziz)
22. "Attempts to reproduce large pre-trained speech models on an academic computing scale," Music and Audio Workshop, June 2023
23. "Attempts to reproduce large pre-trained speech models on an academic computing scale," panel talk at the IEEE ICASSP 2023 workshop Self-supervision in Audio, Speech and Beyond (SASB), June 2023
24. "Compositional framework for spoken language processing," Amazon Alexa Academy Distinguished Speaker Series, May 2023 (host: Jing Liu)
25. "CHiME Speech Separation and Recognition Challenge," ASJ Special Session, March 2023
26. "Controllable and Explainable End-to-End Speech Translation," Tencent, Dec 2022 (host: Dr. Raymond Yu)

27. "Controllable and Explainable End-to-End Speech Translation," SIG SLT Seminar, November 2022
28. "Explainable End-to-End Neural Networks for Far-Field Conversation Recognition," UT Austin, November 2022 (host: Prof. David Harwath)
29. "Toward End-to-End Neural Modeling for Multi-Speaker Conversation Recognition," AIRS Company, Hyundai Motor Group, September 2022 (host: Dr. Byeong-Yeol Kim)
30. "Toward End-to-End Neural Modeling for Multi-Speaker Conversation Recognition," Meta, August 2022
31. "Streaming Transformer for Speech Recognition, Understanding, and Translation," Nvidia, June 2022 (host: Dr. Boris Ginsburg)
32. "Toward End-to-End Neural Modeling for Multi-Speaker Conversation Recognition," Amazon, April 2022 (host: Dr. Tao Zhang)
33. "Toward End-to-End Neural Modeling for Multi-Speaker Conversation Recognition," Pindrop, April 2022 (host: Dr. Ganesh Sivaraman)
34. "End-to-End Speech Recognition and Spoken Language Understanding," JPMC Monthly Speaker Series, February 2022 (host: Dr. Chandra Dhir)
35. "Simplifying Speech Recognition with Non-Autoregressive Modeling," Meta (FB) Speech & Audio Summit, December 2021
36. Panelist of "Convergence of Machine Learning and Signal Processing," APSIPA US Local Chapter, November 2021 (host: Prof. Anthony Kuh)
37. "Multi-Speaker Conversation Recognition based on End-to-End Neural Networks," Georgia Tech CSIP Seminar, November 2021
38. "Simplifying Speech Recognition with Non-Autoregressive Modeling," Amazon Pittsburgh, October 2021 (host: Dr. Markus Müller)
39. "Tackling Multispeaker Conversation Processing based on Speaker Diarization and Multispeaker Speech Recognition," Tencent Seattle AI Lab Online Talk, July 2021 (host: Dr. Meng Yu)
40. "Tackling Multispeaker Conversation Processing based on Speaker Diarization and Multispeaker Speech Recognition," Speech Science Forum at UCL, May 2021
41. "Introduction of ESPnet, end-to-end speech processing toolkit: new features, broadened applications, performance improvements, and future challenges," AIST AI Seminar, March 2021 (host: Dr. Jun Ogata)
42. "Tackling Multispeaker Conversation Processing based on Speaker Diarization and Multispeaker Speech Recognition," CUED Speech Group Seminars, February 2021 (host: Prof. Kate Knill)
43. "Tackling Multispeaker Conversation Processing based on Speaker Diarization and Multispeaker Speech Recognition," Indiana University Bloomington Computer Science Department Colloquium, October 2020 (host: Prof. Donald S. Williamson)
44. "End-to-End Speech Processing: From Pipeline to Integrated Architecture," CMU, USA, October 2019 (host: Prof. Florian Metze)
45. "End-to-End Speech Processing: From Pipeline to Integrated Architecture," IBM, USA, August 2019 (host: Dr. Xiaodong Cui)
46. "End-to-End Speech Processing: From Pipeline to Integrated Architecture," University of Sheffield, UK, May 2019 (host: Prof. Jon Barker)
47. "Multichannel End-to-End Speech Recognition," Toyota Technological Institute at Chicago (TTIC), USA, November 2018 (host: Prof. Karen Livescu)
48. "Multichannel End-to-End Speech Recognition," Paderborn University, Germany, October 2018 (host: Prof. Reinhold Häb-Umbach)
49. "Hybrid CTC/Attention Architecture for End-to-End Speech Recognition," National Chiao Tung University, Taiwan, October 2018 (host: Prof. Jen-Tzung Chien)
50. "Multichannel End-to-End Speech Recognition," Academia Sinica, Taiwan, October 2018 (host: Prof. Yu Tsao)
51. "Distant-Talk Speech Processing Toward Natural Conversation Understanding," Johns Hopkins University, USA, January 2017 (host: Prof. Najim Dehak)

52. "Pushing the envelope at both ends — beamforming acoustic models and joint CTC/Attention schemes for end-to-end ASR," Tokyo Institute of Technology, Japan, December. 2016 (host: Prof. Koichi Shinoda)
53. "Recent activities of distant talk speech recognition," Nanyang Technological University, Singapore, March. 2016 (host: Dr. Xiong Xiao)
54. "Recent activities of distant talk speech recognition," Shanghai Jiao Tong University, China, March. 2016 (host: Prof. Kai Yu)
55. "Practical Bayesian methods for speech and language processing," Brno University of Technology, Czech Republic, September. 2015 (host: Prof. Jan "Honza" Cernocky)
56. "Recent trends in far-field speech recognition," Nara Institute of Science and Technology, Japan, March. 2015 (host: Profs. Kevin Duh and Satoshi Nakamura)
57. "Recent trends on noise robust speech recognition," Tokyo Institute of Technology, Japan, October. 2014 (host: Prof. Takahiro Shinozaki)
58. "Bayesian Learning for Speech and Language Processing," The Institute of Statistical Mathematics, Japan, March. 2013 (host: Prof. Tomoko Matsui)
59. "Bayesian Learning for Speech and Language Processing," Ruhr-Universität Bochum, Germany, Sep. 2013 (host: Prof. Dorothea Kolossa)
60. "Bayesian Learning for Speech and Language Processing," National Taiwan University, Taiwan, July. 2013 (host: Prof. Lin-shan Lee)
61. "Structural Bayesian linear regression for hidden Markov models" Speech Signal Processing Workshop 2013, Taiwan, June. 2013 (host: Prof. Jen-Tzung Chien)
62. The 4th Young Researchers Forum on ALAGIN Speech Processing Session, Japan, Dec. 2012 (host: Prof. Daisuke Saito)
63. "Bayesian Learning for Speech and Language Processing," BBN Technology, USA, Jun. 2012 (host: Dr. Ivan Bulyko)
64. "Speech Recognition by Tracking Acoustical and Linguistic Environments," MIT CSAIL Seminar, USA, April 2012 (host: Prof. Jim Glass)
65. "Bayesian Linear Regression For Hidden Markov Model Based On Optimizing Variational Bounds," NICT, Japan, Nov. 2011 (host: Dr. Chiori Hori)
66. "Bayesian Linear Regression For Hidden Markov Model Based On Optimizing Variational Bounds," Tsinghua University, China, Sep 2011 (host: Prof. Zhijian Ou)
67. "Incremental adaptation of speech recognition based on macroscopic time evolution system," Seminar on Human Language Technology and Pattern Recognition, RWTH Aachen University, Germany, Sep 2011 (host: Prof. Ralf Schlüter)
68. "Incremental adaptation of speech recognition based on macroscopic time evolution system," Seminar on Tokuda & Lee Laboratory, Nagoya Institute of Technology, Japan, June 2011 (host: Prof. Keiichi Tokuda)
69. "Incremental adaptation of speech recognition based on macroscopic time evolution system," Seminar on Machine Intelligence Laboratory, Cambridge University Department of Engineering, UK, May 2011 (host: Dr. Kai Yu)
70. "Speech recognition based on a Bayesian approach," Seminar on NHK Science & Technology Laboratories, Japan, September 2009 (host: Dr. Toru Imai)
71. "Variational Bayesian estimation and clustering for speech recognition," Georgia tech. CSIP seminar, USA, February 2009 (host: Prof. Fred Juang).

PROFESSIONAL SERVICE

MEMBERSHIP

1. Fellow "for contributions to speech recognition technology," the Institute of Electrical and Electronics Engineers (IEEE)
2. Fellow "For wide ranging, fundamental contributions to research and leadership in speech recognition technologies," the International Speech Communication Association (ISCA)
3. Member, the Acoustical Society of Japan (ASJ)
4. Member, the Institute of Electronics, Information, and Communication Engineers of Japan (IEICE)

ORGANIZER/COMMITTEE MEMBER

1. General Chair, IEEE International Conference on Audio, Speech, and Language Intelligence (ASLI) 2027
2. Co-organizer, "SALMA 2026: Speech and Audio Language Models Workshop (2nd Edition)," at EMNLP 2026 Satellite Workshop
3. Lead Area Chair, INTERSPEECH 2025 (2025.10 -- 2027.09)
4. Co-organizer, "The Joint Workshop on HSCMA and CHiME 2026," at ICASSP 2026 Satellite Workshop
5. Co-Organizers & Academic Partners, "The 2025 LRAC Challenge," at ICASSP 2026 Satellite Workshop
6. Challenge co-organizer, "URGENT Challenge Universality, Robustness, and Generalizability of speech EnhancemeNT systems," at ICASSP 2026 Grand Challenge
7. Evaluation Committee for the Best Paper Award of the Speech Communication in 2025
8. Challenge co-organizer, "MISP 2025 challenge," at Interspeech 2025
9. Area Chair, IJCAI-25 (2024.11 -- 2025.08)
10. Chair, the IEEE Signal Processing Society Speech and Language Technical Committee (IEEE SLTC) (2025.1 - 2026.12)
11. Challenge co-organizer, "The ML-SUPERB 2.0 Challenge: Towards Fair and Robust Speech Processing," at Interspeech 2025
12. Challenge co-organizer, "URGENT Challenge Universality, Robustness, and Generalizability of speech EnhancemeNT systems," at Interspeech 2025
13. Lead Area Chair, INTERSPEECH 2025 (2024.09 -- 2025.08)
14. Special session co-organizer, "Self-Supervised Learning for Conversational Speech Processing," at ICASSP 2025
15. General Co-Chair, IEEE ICASSP 2028 (2024.05 -- 2028.05)
16. NCSA's DeltaAI External Advisory Committee (2024.04--)
17. Co-organizer, "Audio Imagination" at NeurIPS 2024 Workshop
18. Co-organizer, "CHiME 2024 Workshop" at Insterspseech Sattelite Workshop 2024
19. Co-organizer, "SynData4GenAI" at Insterspseech Sattelite Workshop 2024
20. Challenge co-organizer, "Speech Processing Using Discrete Speech Units" at Interspeech 2024
21. Special session co-organizer, "Spoken Language Models for Universal Speech Processing" at Interspeech 2024
22. Lead Area Chair, INTERSPEECH 2024 (2023.11 -- 2024.09)
23. Vice chair, the IEEE Signal Processing Society Speech and Language Technical Committee (IEEE SLTC) (2024.1 - 2024.12)
24. Senior Area Chair "Speech and Multimodality" in EMNLP 2023
25. ISCA Fellows Selection Committee (2023--24)
26. Area Chair "Speech and Multimodality" in ACL 2023
27. Co-Area Chair, INTERSPEECH 2023 (2022.10 -- 2023.08)
28. Member, the IEEE Signal Processing Society Speech and Language Technical Committee (IEEE SLTC) (2023.1 - 2025.12)
29. Organizing Committee, SANE 2022 - Speech and Audio in the Northeast (2022.10)
30. Special session co-organizer, "Resource-efficient Real-time Neural Speech Separation," at ICASSP 2023
31. International Liaison Committee for 2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU) (2022.9 - 2023.12)
32. TPC Co-Chairs, 14th Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC) (2022.5 -- 2022.11)
33. Area Chair, 2022 International Workshop on Acoustic Signal Enhancement (IWAENC 2022) (2022.5-2022.10)
34. Technical Chairs, 2022 IEEE Spoken Language Technology Workshop (SLT 2022) (2021.10 - 2023.1)
35. Publication Chair, Interspeech 2024
36. Special session co-organizer, "Non-Autoregressive Sequential Modeling for Speech Processing" at Interspeech 2021
37. Special session co-organizer, "Far-field Multi-Channel Speech Enhancement Challenge for Video Conferencing (ConferencingSpeech 2021)" at Interspeech 2021

38. TPC Chairs, 13th Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC) (2020.4 -- 2020.12)
39. Co-organizer, NeurIPS 2020 workshop "Self-Supervised Learning for Speech and Audio Processing"
40. Area Chair, IEEE Spoken Language Technology Workshop (IEEE SLT 2021) (2020.8 -- 2021.1)
41. Senior Program Committee Member, the Thirty-Fifth AAAI Conference on Artificial Intelligence (AAAI 2021) (2020.8 -- 2021.2)
42. Area Chair, the 30th International Joint Conference on Artificial Intelligence (IJCAI 2021) (2020.8 -- 2021.4)
43. Senior Program Committee Member, the 29th International Joint Conference on Artificial Intelligence and the 17th Pacific Rim International Conference on Artificial Intelligence (IJCAI-PRICAI 2020) (2019.11 -- 2020.7)
44. Area Chair, the Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI-20) (2019.8 -- 2020.2)
45. APSIPA Distinguished Lecturers (2019-2020)
46. IEEE Signal Processing Society Liaison for 2019 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU) (2018.9 - 2019.12)
47. Organizing Committee, 2018 SANE 2018 - Speech and Audio in the Northeast (2018.5-2018.10)
48. Member, the IEEE Signal Processing Society Machine Learning for Signal Processing Technical Committee (IEEE MLSP) (2018.1 - 2020.12)
49. Senior Member, Co-Leader, "Multilingual End-to-end ASR for Incomplete Data" at 2018 Jelinek Summer Workshop on Speech and Language Technology (2017.12--2018.8)
50. Area Chair "Vision, Robotics, Multimodal, Grounding and Speech" in ACL 2018
51. Special session co-organizer, "Multi-Microphone Speech Recognition" at ICASSP 2018.
52. Program Chair, the 2017 IEEE International Workshop on Machine Learning for Signal Processing (MLSP'17) (2016.9 - 2017.9)
53. Technical Chair, IEEE ASRU 2017 (2016.5--2017.12)
54. Organizing Committee, the 4th International Workshop on Speech Processing in Everyday Environments (CHiME 2016), (2016.5 - 2016.9)
55. Senior Affiliates, "Building Speech Recognition System from Untranscribed Data" at 2016 Jelinek Summer Workshop on Speech and Language Technology (2015.12--2016.8)
56. Chair, Industrial Membership Committee, APSIPA (2016-)
57. Organizing Committee, the 4th CHiME Speech Separation and Recognition Challenge (2016.4-2016.9)
58. Member, IEEE CIS Task Force on Computational Audio Processing (2015.12-)
59. Interactive Peer Reviewer, the 2016 Frederick Jelinek Memorial Summer Workshop on Speech and Language Technology (JSALT 2016), (2015.11)
60. Senior Member, "Far-Field Speech Enhancement and Recognition in Mismatched Settings" at 2015 Jelinek Summer Workshop on Speech and Language Technology (2014.12--2015.8)
61. Special session co-organizer, "Robust Speech Processing using Observation Uncertainty and Uncertainty Propagation" at Interspeech (2015)
62. Organizing Committee, The 3rd CHiME Speech Separation and Recognition Challenge at IEEE ASRU 2015 (2014.7-2015.12)
63. Sponsorship Chair, IEEE ASRU 2015 (2014.6-2015.12)
64. TPC member, Machine Learning Applications in Speech Processing Symposium in the 2nd IEEE Global Conference on Signal and Information Processing (Global SIP) (2013.12 - 2014.12)
65. Industrial Membership Committee, APSIPA (2014.1 - 2017.12)
66. Member, the APSIPA Speech, Language, and Audio Technical Committee (APSIPA SLA TC) (2014.1 - 2019.12)
67. Member, the IEEE Signal Processing Society Speech and Language Technical Committee (IEEE SLTC) (2014.1 - 2019.12)
68. Organizing Committee, the 2012 SANE 2012 - Speech and Audio in the Northeast (2012.7-2012.10)
69. Organizing Committee, the 2012 IEEE AASP 'CHiME' SpeechSeparation and Recognition Challenge (2012.3-2013.3)
70. Publication Chair, the 2012 International Workshop on Statistical Machine Learning for Speech Processing (IWSML2012)
71. Executive Committee Member, the 2nd Young Researchers Forum on ALAGIN Speech Processing Session (2011.2 - 2011.3)

72. Panelist in IPSJ/2009-SLP-79, Tokyo, Japan (2009.12)
73. Program Committee Member, the 12th Workshop on Information-Based Induction Sciences (IBIS 2009) (2008.12 - 2009.11)

EDITOR

1. Associate Editor, EURASIP Signal Processing Open (2026.06 --)
2. Guest Editor, IEEE JSTSP Special issue on Deep Multimodal Speech Enhancement and Separation (2024.10 -- 2025.3)
3. Lead Guest Editor, Elsevier Computer Speech and Language (2024.8 --)
4. Senior Area Editors, IEEE/ACM Transactions on Audio Speech and Language Processing (2022.5 -- 2025.6)
5. Guest Editor, IEEE JSTSP Special issue on Self-Supervised Learning for Speech and Audio Processing (2021.4 -- 2022.1)
6. Guest Editor, Elsevier Computer Speech and Language (2020.10 -- 2021.10)
7. Subject Editor, Elsevier Speech Communication (2019.4 --)
8. Lead Guest Editor, IEEE JSTSP Special issue on Far-Field Speech Processing in the Era of Deep Learning (2018.5 -- 2019.8)
9. Associate Editor, APSIPA Transactions on Signal and Information Processing (2018.4 --)
10. Associate Editor, IEICE Transactions on Information and Systems (2016.7 -- 2020)
11. Guest Editor, Elsevier Computer Speech and Language (2015.12 -- 2016.11)
12. Guest Associate Editor, IEICE Transactions on Information and Systems (2013.4 -- 2014.6)
13. Associate Editor, IEEE Transactions on Audio, Speech, and Language Processing (ASLP). (2012.4 -- 2015.4)

SESSION CHAIR

1. IEEE MLSP (2017)
2. ISCA Interspeech (2015, 2016, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025)
3. IEEE ICASSP (2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026)
4. IEEE ASRU (2015, 2017, 2021, 2025)
5. IEEE SLT (2021, 2022)
6. Acoustical Society of Japan Meeting (2004.3-2010)

CONFERENCE REVIEW

1. Asian Conference on Machine Learning (ACML)
2. Association for Computational Linguistics (ACL)
3. Conference on Language Modeling (COLM)
4. DCASE workshop
5. Empirical Methods in Natural Language Processing (EMNLP)
6. European Chapter of the Association for Computational Linguistics (EACL)
7. European Conference on Computer Vision (ECCV)
8. European Signal Processing Conference (EUSIPCO)
9. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
10. IEEE Global Conference on Signal and Information Processing (GlobalSIP)
11. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)
12. IEEE International Workshop on Machine Learning for Signal Processing (MLSP)
13. IEEE Workshop on Automatic Speech Recognition and Understanding (ASRU)
14. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
15. International Conference on Learning Representations (ICLR)
16. International Conference on Machine Learning (ICML)
17. International Conference on Pattern Recognition (ICPR)
18. International Conference on Artificial Neural Networks (ICANN)
19. International Joint Conferences on Artificial Intelligence (IJCAI)

20. International Joint Conference on Neural Networks (IJCNN)
21. International Symposium on Chinese Spoken Language Processing (ISCSLP)
22. International Workshop on Acoustic Signal Enhancement (IWAENC)
23. International Workshop on Statistical Machine Learning for Speech Processing (IWSML)
24. ISCA INTERSPEECH
25. ISCA Young Female Researchers in Speech Workshop (YFRSW)
26. Joint Workshop on Hands-free Speech Communication and Microphone Arrays (HSCMA)
27. Neural Information Processing Systems (NeurIPS)
28. REVERB workshop

JOURNAL/TRANSACTION REVIEW

1. APSIPA Transactions on Signal and Information Processing
2. Computer Speech and Language
3. EURASIP Journal on Advances in Signal Processing
4. IEEE Journal of Selected Topics in Signal Processing
5. IEEE Signal Processing Letters
6. IEEE Signal Processing Magazine
7. IEEE Transaction on Audio, Speech and Language Processing
8. IEEE Transaction on Emerging Topics in Computational Intelligence
9. IEEE Transaction on Signal Processing
10. Information Processing Society of Japan (IPSJ) Transactions
11. Institute of Electronics, Information and Communication Engineers (IEICE) Transactions
12. Journal of Signal Processing Systems
13. Journal of Statistical Planning and Inference
14. Journal of the Acoustical Society of America
15. Journal of the Acoustical Society of Japan
16. Speech Communication
17. Transactions of the Association for Computational Linguistics (TACL)
18. Transactions on Machine Learning Research (TMLR)

EXTERNAL PHD THESIS REVIEW/COMMITTEE

1. Ryan Whetten (University of Avignon)
2. Pu Wang (KU Leuven)
3. Sreyan Ghosh (University of Maryland)
4. Kohei Saijo (Waseda University)
5. Julius Richter (University of Hamburg)
6. Salah Zaiem (IP Paris)
7. Wen-Chin Huang (Nagoya University)
8. Yusuke Fujita (Waseda University)
9. Yosuke Higuchi (Waseda University)
10. Salvador Medina Maza (CMU)
11. Billy Li (CMU)
12. Ngoc Quan Pham (Karlsruhe Institute of Technology)
13. Albert Zeyer (RWTH Aachen University)
14. Chunting Zhou (CMU)
15. Shucong Zhang (University of Edinburgh)
16. Thai Son Nguyen (Karlsruhe Institute of Technology)

17. Ruizhi Li (JHU)
18. Nelson Enrique Yalta Soplin (Waseda University)
19. Suyoun Kim (CMU)
20. Tomoki Hayashi (Nagoya University)
21. Dung Tien Tran (INRIA)
22. Antti Hurmalainen (Tampere University of Technology)

RESEARCH PROJECTS

1. Anuttacon (2025 –)
2. Sesami (2025 –)
3. DSTA (2024 –)
4. Amazon (2024 –)
5. Apple (2023.5--)
6. NAVER/LINE (2021.12--2022.11)
7. Honda Research Institute, Japan (2021.9--2025)
8. JHU HLTCOE (2021.9--)
9. Google, USA (2021.9--2022.11)
10. Facebook, USA (2021.9--2024)
11. Hyundai, Korea (2021.7--2024)
12. MIT Lincoln Laboratory, USA (2021.2--)
13. ASAPP, USA (2020.5--2024)
14. NAVER, Korea (2020.5--2021.4)
15. Sony, Tokyo, Japan (2019.7--)
16. Yahoo! Japan, Tokyo, Japan (2018.9--)
17. Hitachi Corporation, Tokyo, Japan (2018.7--2023)
18. NTT Communication Science Laboratories, Kyoto, Japan. (2017.12 --)
19. Professor Takahiro Shinozaki at Tokyo Institute of Technology, Tokyo, Japan. (2014.4 – 2018.3)
20. Professor Shigeki Sagayama at the University of Tokyo, Tokyo, Japan. (2008.7 – 2011.11)
21. Professor Tetsunori Kobayashi at Waseda University, Tokyo, Japan. (2009.9 – 2010.2)
22. Professor Shigeru Katagiri at Doshisha University, Kyoto, Japan. (2010. 4 – 2011.11)
23. Professor Nobuyuki Minematsu at the University of Tokyo, Tokyo, Japan. (2010.7 – 2011.3)
24. Professor Biing-Hwang (Fred) Juang at Georgia Institute of Technology, Atlanta, USA. (2010.11 – 2011.11)

ADVISING AND COLLABORATION

AT CMU

1. 2023. 09 - 2024. 06: Karen Livescu (TTI-Chicago)
2. 2021. 08 - 2024. 07: Zhong-Qiu Wang
3. 2022. 03 - 2024. 08: Soumi Maiti
4. 2023. 05 - 2024. 09: Jee-Weon Jung
5. 2023. 10 -: Samuele Cornell
6. 2024. 02 - 2025. 05: Hye-jin Shim
7. 2025. 08 -: Jaeyeon Kim (co-supervisor)
8. 2025. 08 -: Chien-yu Huang
9. 2024. 08 -: Shikhar Bharadwaj
10. 2024. 08 - 2026. 05: Chyi-Jiunn Lin

11. 2024. 08 -: Masao Someki
12. 2023. 08 -: 2025. 05: Kwanghee Choi
13. 2023. 08 -: Jinchuan Tian
14. 2022. 08 - 2024. 05: Shih-Lun Wu
15. 2022. 08 -: William Chen
16. 2022. 05 - 2026. 04: Li-wei Chen (co-supervisor)
17. 2022. 04 - 2024. 01: Jessica Huynh (co-supervisor)
18. 2022. 02 - 2024. 06: Muqiao Yang (co-supervisor)
19. 2021. 09 - 2025. 05: Yifan Peng
20. 2021. 09 - 2023. 08: Dan Berrebbi
21. 2021. 08 - 2023. 05: Xinjian Li
22. 2021. 08 - 2026. 01: Jiatong Shi
23. 2021. 05 - 2026. 03: Siddhant Arora
24. 2021. 01 - 2024. 06: Xuankai Chang
25. 2020. 09 - 2026. 05: Brian Yan
26. 2022. 01 - 2022. 12: Dorsa Zeinali
27. 2021. 09 - 2022. 12: Karthik Ganesan (MIIS directed study)
28. 2022. 01 - 2022. 08: Debayan Ghosh
29. 2021. 09 - 2022. 08: Sujay Suresh Kumar (MIIS directed study)
30. 2021. 01 - 2022. 08: Peter Wu (co-supervisor)
31. 2021. 01 - 2022. 08: Chaitanya Narisetty
32. 2020. 09 - 2022. 08: Siddharth Dalmia (co-supervisor)
33. 2026. 05 - 2027. 05: Kuang-Da Wang (National Yang Ming Chiao Tung University)
34. 2026. 05 - 2026. 08: Thanapat Trachu (University of Southern California)
35. 2026. 02 - 2026. 07: Dahee Yang (Hanyang University)
36. 2026. 01 - 2026. 05: Alexander Polok (Brno University of Technology)
37. 2025. 12 - 2026. 06: Xun Gong (Shanghai Jiao Tong University)
38. 2025. 08 - 2025. 12: Haoran Wang (Shanghai Jiao Tong University)
39. 2025. 05 - 2025. 11: Ji Hoon Kim (KAIST)
40. 2025. 04 - 2026. 03: Bo-Hao Su (National Tsing Hua University)
41. 2025. 02 - 2025. 07: Jialu Li (UIUC)
42. 2025. 01 - 2025. 04: Pu Wang (KU Leuven)
43. 2024. 10 - 2024. 12: Carlos Carvalho (Instituto Superior Técnico (IST))
44. 2024. 10 - 2025. 01: Holger Severin Bovbjerg (Aalborg University)
45. 2024. 11 - 2024. 12: Junyi Peng (Brno University of Technology)
46. 2024. 08 - 2024. 12: Kalvin Chang (CMU)
47. 2024. 08 - 2024. 11: Yoshiaki Band (AIST)
48. 2024. 07 - 2024. 11: Shih-Heng (Stan) Wang (National Taiwan University)
49. 2024. 04 - 2024. 12: Shuichiro Shimizu (Kyoto University)
50. 2023. 11 - 2024. 04: Chenda Li (Shanghai Jiao Tong University)
51. 2023. 09 - 2024. 08: Yihan Wu (Renmin University)
52. 2023. 08 - 2023. 11: Min Su Kim (KAIST)
53. 2023. 04 - 2023. 07: Kohei Saijo (Waseda University)
54. 2023. 02 - 2024. 02: Wangyou Zhang (Shanghai Jiao Tong University)
55. 2022. 11 - 2023. 01: Takaaki Saeki (University of Tokyo)

56. 2022. 04 - 2022. 09: Samuele Cornell (Universita Politecnica delle Marche)
57. 2022. 03 - 2022. 06: YoshikiMasuyama (Tokyo Metropolitan University)
58. 2022. 03 - 2022. 06: Yosuke Higuchi (Waseda University)
59. 2021. 12 - 2022. 12: Yosuke Kashiwagi (Sony)
60. 2021. 08 - 2021. 12: Yen-Ju Lu (Academia Sinica)
61. 2021. 07 - 2022. 07: Yushi Ueda (Japan Patent Office)

AT JHU

1. 2020. 08 - 2021. 08: Tianzi Wang (CUHK)
2. 2019. 09 - 2021. 09: Jiatong Shi (transferred to CMU)
3. 2019. 09 - 2020. 12: Xuankai Chang (transferred to CMU)
4. 2018. 07 - 2019. 06: Zhiqi Wang
5. 2017. 12 - 2020. 12: Matthew Wiesner (co-supervisor, JHU HLTCOE)
6. 2017. 10 - 2022. 05: Matthew Maciejewski (co-supervisor, Amazon)
7. 2017. 09 - 2018. 12: Szu-Jui Chen
8. 2017. 09 - 2022. 05: Aswin Shanmugam Subramanian
9. 2020. 01 - 2021. 01: Pengcheng Guo (Northwestern Polytechnical University)
10. 2019. 12 - 2020. 03: Yosuke Higuchi (Waseda University)
11. 2019. 12 - 2020. 12: Jing Shi (Chinese Academy of Science)
12. 2019. 07 - 2019. 10: Katsuki Inoue (Okayama University)
13. 2018. 11 - 2019. 05: Murali Karthick Baskar (Brno University of Technology)
14. 2018. 09 - 2018. 12: Xuankai Chang (Shanghai Jiao Tong University)
15. 2018. 08 - 2018. 09: Hirofumi Inaguma (Kyoto University)
16. 2018. 07 - 2020. 03: Yusuke Fujita (Hitachi Ltd.)
17. 2018. 04 - 2018. 09: Nelson Enrique Yalta Soplin (Waseda University)

AT MERL

1. 2016. 11 - 2016. 01: Tsubasa Ochiai (Doshisha University → NTT Labs)
2. 2016. 08 - 2016. 11: Tomoki Hayashi (Nagoya University)
3. 2016. 05 - 2016. 08: Zhong Meng (Georgia Institute of Technology → Microsoft)
4. 2016. 05 - 2016. 08: Suyoun Kim (Carnegie Mellon University → Facebook)
5. 2015. 09 - 2015. 12: Morteza Shahriari (University of Florida)
6. 2014. 11 - 2015. 05: Zhuo Chen (Columbia University → Microsoft)
7. 2014. 09 - 2014. 11: Ahmed Hussen Abdelaziz (Ruhr-Universität Bochum → ICSI → Apple)
8. 2014. 06 - 2014. 09: Yi Luan (University of Washington)
9. 2013. 09 - 2014. 02: Felix Weninger (Technische Universität München → Nuance)
10. 2013. 06 - 2013. 08: Hao Tang (Toyota Technological Institute at Chicago → MIT → University of Edinburgh)
11. 2012. 12 - 2013. 02: Koichiro Yoshino (Kyoto University → NAIST)

AT NTT

1. 2011. 05 - 2011. 08: Ekapol Chuangsuwanich (Massachusetts Institute of Technology → Chulalongkorn University)
2. 2011. 01 - 2011. 03: Yasuhisa Fujii (Toyohashi University of Technology → Google)
3. 2010. 01 - 2010. 08: Jose Domingo Esparza Garcia (Technical University of Cartagena → Ulm University)
4. 2010. 01 - 2010. 03: Daisuke Saito (University of Tokyo)
5. 2009. 08: Denis Babani (NAIST → OwnerIQ, Inc.)
6. 2009. 08 - 2009. 09, 2010. 08 - 2010. 09: Naohiro Tawara (Waseda University → NTT Labs)
7. 2008. 10 - 2008. 12: Yotaro Kubo (Waseda University → NTT Labs → Amazon → Google)

8. 2007. 12 - 2008. 11: David Cournapeau (Kyoto University → Enthought)
9. 2007. 08: Kenta Nishiki (University of Tokyo → NTT Labs)
10. 2006. 01 - 2006. 08: Rasa Narkeviciute (Kaunas University of Technology → AEA Technology)
11. 2006. 01 - 2006. 02: Hirokazu Kameoka (University of Tokyo → NTT Labs/University of Tokyo)
12. 2005. 07 - 2006. 05: Wesley William Arnquist (University of Washington → Costco Wholesale)
13. 2003. 08: Toshiaki Kubo (Waseda University → Mitsubishi Electric)
14. 2003. 04 - 2004. 03, 2008.10: Atsushi Sako (Ryukoku University → Kobe University → Nintendo)

PUBLICATIONS

BOOK

1. **Shinji Watanabe**, Marc Delcroix, Florian Metze, and John R. Hershey, "New Era for Robust Speech Recognition -Exploiting Deep Learning-," *Springer* (2017)
2. **Shinji Watanabe** and Jen-Tzung Chien, "Bayesian Speech and Language Processing," *Cambridge University Press* (2015)

BOOK CHAPTER

1. Takahiro Shinozaki, **Shinji Watanabe**, Kevin Duh, "Automated Development of DNN Based Spoken Language Systems Using Evolutionary Algorithms," in "Deep Neural Evolution: Deep Learning with Evolutionary Computation," eds., Hitoshi Iba, Nasimul Noman, *Springer*, pp. 97--129, (2020)
2. **Shinji Watanabe**, Tuomas Virtanen, Dorothea Kolossa, "Application of Source Separation to Robust Speech Analysis and Recognition," in "Audio Source Separation and Speech Enhancement," eds. Emmanuel Vincent, Tuomas Virtanen, Sharon Gannot, *John Wiley & Sons Ltd* (2018)
3. Marc Delcroix, **Shinji Watanabe**, and Tomohiro Nakatani, "Variance Compensation for Recognition of Reverberant Speech with Dereverberation Preprocessing, " in "Robust Speech Recognition of Uncertain or Missing Data, " eds. Dorothea Kolossa and Reinhold Haeb-Umbach, *Springer*, pp. 225--256, (2011)

PHD THESIS

1. Speech recognition based on a Bayesian approach, Waseda University (2006)

REVIEW AND OVERVIEW PAPER

1. Samuele Cornell, Christoph Boeddeker, Taejin Park, He Huang, Desh Raj, Matthew Wiesner, Yoshiki Masuyama, Xuankai Chang, Zhong-Qiu Wang, Stefano Squartini, Paola Garcia, and **Shinji Watanabe**, "Recent trends in distant conversational speech recognition: A review of CHiME-7 and 8 DASR challenges," *Computer Speech & Language*, vol 97, page 101901 (2026)
2. Siddhant Arora, Kai-Wei Chang, Chung-Ming Chien, Yifan Peng, Haibin Wu, Yossi Adi, Emmanuel Dupoux, Hung-yi Lee, Karen Livescu, and **Shinji Watanabe**, "On The Landscape of Spoken Language Models: A Comprehensive Survey," *Transactions on Machine Learning Research* (2025)
3. Pooneh Mousavi, Gallil Maimon, Adel Moumen, Darius Petermann, Jiatong Shi, Haibin Wu, Haici Yang, Anastasia Kuznetsova, Artem Ploujnikov, Ricard Marxer, Bhuvana Ramabhadran, Benjamin Elizalde, Loren Lugosch, Jinyu Li, Cem Subakan, Phil Woodland, Minje Kim, Hung-yi Lee, **Shinji Watanabe**, Yossi Adi, and Mirco Ravanelli, "Discrete Audio Tokens: More Than a Survey!" *Transactions on Machine Learning Research* (2025)
4. Xuankai Chang, **Shinji Watanabe**, Marc Delcroix, Tsubasa Ochiai, Wangyou Zhang, and Yanmin Qian, "Module-based End-to-End Distant Speech Processing: a case study of far-field ASR," *IEEE Signal Processing Magazine*, vol. 41, no. 6, pp. 39-50 (2024)
5. Rohit Prabhavalkar, Takaaki Hori, Tara N Sainath, Ralf Schlüter, and **Shinji Watanabe**, "End-to-End Speech Recognition: A Survey," *IEEE Transactions on Audio, Speech and Language Processing*, vol. 32, pp. 325-351 (e)2023
6. Abdelrahman Mohamed, Hung-yi Lee, Lasse Borgholt, Jakob D. Havtorn, Joakim Edin, Christian Igel, Katrin Kirchhoff, Shang-Wen Li, Karen Livescu, Lars Maaløe, Tara N. Sainath, and **Shinji Watanabe**, "Self-Supervised Speech Representation Learning: A Review," *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1179-1210 (2022)

7. Antonios Anastasopoulos, Loïc Barrault, Luisa Bentivogli, Marcely Zanon Boito, Ondřej Bojar, Roldano Cattoni, Anna Currey, Georgiana Dinu, Kevin Duh, Maha Elbayad, Clara Emmanuel, Yannick Estève, Marcello Federico, Christian Federmann, Souhir Gahbiche, Hongyu Gong, Roman Grundkiewicz, Barry Haddow, Benjamin Hsu, Dávid Javorský, Věra Kloudová, Surafel Lakew, Xutai Ma, Prashant Mathur, Paul McNamee, Kenton Murray, Maria Nădejde, Satoshi Nakamura, Matteo Negri, Jan Niehues, Xing Niu, John Ortega, Juan Pino, Elizabeth Salesky, Jiatong Shi, Matthias Sperber, Sebastian Stüker, Katsuhito Sudoh, Marco Turchi, Yogesh Virkar, Alexander Waibel, Changhan Wang, **Shinji Watanabe**, "Findings of the IWSLT 2022 Evaluation Campaign," *Proc. IWSLT'22*, pp. 98–157 (2022)
8. Joseph Turian, Jordie Shier, Humair Raj Khan, Bhiksha Raj, Björn W. Schuller, Christian J. Steinmetz, Colin Malloy, George Tzanetakis, Gissel Velarde, Kirk McNally, Max Henry, Nicolas Pinto, Camille Noufi, Christian Clough, Dorien Herremans, Eduardo Fonseca, Jesse Engel, Justin Salamon, Philippe Esling, Pranay Manocha, **Shinji Watanabe**, Zeyu Jin, and Yonatan Bisk, "HEAR 2021: Holistic Evaluation of Audio Representations," *In NeurIPS 2021 Competitions and Demonstrations Track* (pp. 125--145). PMLR (2021)
9. **Shinji Watanabe**, Florian Boyer, Xuankai Chang, Pengcheng Guo, Tomoki Hayashi, Yosuke Higuchi, Takaaki Hori, Wen-Chin Huang, Hirofumi Inaguma, Naoyuki Kamo, Shigeki Karita, Chenda Li, Jing Shi, Aswin Shanmugam Subramanian, and Wangyou Zhang, "The 2020 ESPnet update: new features, broadened applications, performance improvements, and future plans," *Proc. 2021 IEEE Data Science and Learning Workshop (DSLW 2021)*, pp. 1--6 (2021)
10. **Shinji Watanabe**, Michael Mandel, Jon Barker, Emmanuel Vincent, Ashish Arora, Xuankai Chang, Sanjeev Khudanpur, Vimal Manohar, Daniel Povey, Desh Raj, David Snyder, Aswin Shanmugam Subramanian, Jan Trmal, Bar Ben Yair, Christoph Boeddeker, Zhaoheng Ni, Yusuke Fujita, Shota Horiguchi, Naoyuki Kanda, Takuya Yoshioka, and Neville Ryant, "CHiME-6 Challenge: Tackling Multispeaker Speech Recognition for Unsegmented Recordings," *Proc. CHiME 2020*, pp. 1--7 (2020)
11. Reinhold Haeb-Umbach, Jahn Heymann, Lukas Drude, **Shinji Watanabe**, Marc Delcroix, and Tomohiro Nakatani, "Far-Field Automatic Speech Recognition," *Proceedings of the IEEE*, vol. 109.2: pp. 124--148 (2020)
12. Reinhold Haeb-Umbach, **Shinji Watanabe**, Tomohiro Nakatani, Michiel Bacchiani, Björn Hoffmeister, Michael L. Seltzer, Heiga Zen, and Mehrez Souden, "Speech Processing for Digital HomeAssistants," *IEEE Signal Processing Magazine*, vol. 36, no. 6, pp. 111--124 (2019)
13. Takahiro Shinozaki and **Shinji Watanabe**, "Automatic speech recognition and black-box optimization," *The Journal of the Acoustical Society of Japan*, vol. 72, number 10, pp. 644--652, (2016). (in Japanese)
14. Felix Weninger, Hakan Erdogan, **Shinji Watanabe**, Emmanuel Vincent, Jonathan Le Roux, John R. Hershey, and Bjoern Schuller, "Speech enhancement with LSTM recurrent neural networks and its application to noise-robust ASR," *Proc. Latent Variable Analysis and Signal Separation*, pp. 91--99 (2015)
15. **Shinji Watanabe** and Atsushi Nakamura, "Bayesian Approaches to Acoustic Modeling: A Review," *APSIPA Transactions on Signal and Information Processing*, Volume 1, e5 (11pages), (2012)
16. Mark Gales, **Shinji Watanabe**, Eric Fossler-Lussier, "Structured Discriminative Models for Speech Recognition," *IEEE Signal Processing Magazine*, vol. 29, no.6, pp. 70--82 (2012).
17. **Shinji Watanabe** and Atsushi Nakamura, "Tutorial: Discriminative Training in Speech Recognition," (in Japanese) *The Journal of the Institute of Electronics, Information and Communication Engineers (IEICE)*, vol. 94(10), pp. 920--922, (2011)
18. **Shinji Watanabe**, "Acoustic models in speech recognition," *The Journal of the Acoustical Society of Japan*, vol. 66, number 1, pp. 18--22, (2010). (in Japanese)
19. Atsushi Nakamura, **Shinji Watanabe**, Takaaki Hori, Erik McDermott, and Shigeru Katagiri, "Advanced Computational Models and Learning Theories for Spoken Language Processing," *IEEE Computational Intelligence Magazine*, vol. 1, issue 2, pp. 5--9, (2006)
20. **Shinji Watanabe**, "Speech recognition based on a Bayesian approach," *The Journal of the Acoustical Society of Japan*, vol. 62, number 8, pp. 599--604, (2006). (in Japanese)

JOURNAL (REFEREED)

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2. Hiroaki Sato, Asahi Sakuma, Ryuga Sugano, Tadashi Kumano, Yoshihiko Kawai, **Shinji Watanabe**, and Tetsuji Ogawa, "Uncertainty-Based Streaming ASR with Evidential Deep Learning," *IEEE Open Journal of Signal Processing* (accepted)

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